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Hourly Wildfire Spread Prediction Model

Using Neural Networks

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Abstract

Understanding wildfire spread is a critical environmental and public safety concern. Current machine learning models primarily operate at a daily temporal resolution, but relevant data is often available on hourly scales. To address this, we propose a custom spatiotemporal convolutional neural network that predicts hourly wildfire spread at a 2km spatial resolution. The architecture of the model combines a convolutional gated recurrent unit (ConvGRU), an encoder-decoder structure, and a convolutional block attention module (CBAM) applied to skip connections. To improve predictive performance and physical consistency, we introduce a hybrid loss function consisting of Focal Loss, Focal-Tversky Loss, and a physics-informed regularizer based on a wind-advection partial differential equation. Using 38 environmental, meteorological, and terrain variables, we trained the model on eight California wildfires (3560 hours, 2.9 million acres) and evaluated it on four held-out test fires (1809 hours, 726 thousand acres). The model achieves an aggregate F2 score of 0.928 on the test set, with precision of 0.922 and recall of 0.930.

Introduction

Wildfires are among the most destructive natural disasters that currently affect the United States. Due to their increasing frequency, severity, and socioeconomic impact, accurate prediction of wildfire spread is essential for resource allocation, evacuation planning, and hazard mitigation. According to the National Interagency Fire Center, there were 77,850 fires in 2025 that burned a total of 5,131,474 acres of land. The Los Angeles fires in January 2025 cost \$61.2 billion alone (Climate Center). This highlights the need for more timely and precise predictive models.

Rothermel's 1972 paper *A Mathematical Model for Predicting Fire Spread in Wildland Fuels* set out the currently used equations for modeling and predicting the spread of wildfires (Finney). This approach is largely based on physics-driven formulations, but it is limited in accuracy due to simplifying assumptions, and the difficulty of fully representing the complex interactions among fuels, terrain, and fire dynamics.

In recent years, machine-learning approaches, including NWDS (Huot et al., 2022), WildfireSpreadTS (Gerard et al., 2023), and WSTS+ (2025), have now framed fire spread as an image-prediction problem. These models demonstrate the potential of data-driven approaches to learn fire dynamics directly from observations. However, every such published model has operated at a daily temporal resolution, while most applicable variables are available at an hourly resolution. This leaves a gap where no ML model predicts fire spread at a sub-daily frequency.

In this work, we develop a data-driven model for high-resolution, hourly wildfire spread prediction from the preceding six hours at 2km spatial resolution. The model couples

convolutional and recurrent components to capture patterns at both fine and broad spatial scales and to retain temporal memory across the six-hour input window. It processes 38 input variables spanning fire state, hourly and daily meteorology, and static terrain variables. We combine Focal and Focal-Tversky loss to downweight easy negatives and heavily penalize false negatives, reflecting the greater cost of missing fire spread than overpredicting it. Additionally we include a physics-informed regularizer to constrain predictions to physically plausible wind-driven spread behavior. The following sections describe the dataset, model architecture, and training procedure, followed by an evaluation on four held-out California wildfires.

Data Sets

The model uses a number of variables and datasets to predict fire spread behavior. Meteorological data is collected from local weather stations and includes temperature, wind speed, wind direction, and precipitation. Fire data is taken from GRIDMET and uses previously established variables used to measure fire spread and fuel quality such as ERC, BI, FM100 and FM1000. Energy release component (ERC) is used as a standard measure of fuel quality that takes into account the amount of vegetation, amount of dead and live material, and the moisture in the area. Burn Index (BI) combines ERC with the spread component which measures the predicted speed of a fire. Finally, FM100 and FM1000 measure fuel moisture by accounting for precipitation in the previous 100 and 1,000 hours respectively. All of the variables relating to fire data are updated daily. Topological data is also collected such as slope, aspect, and average elevation of the area. A full list of all variables, sources for variables, and level of detail of the data can be found in Table 1.

The 38 channels were selected through ablations, run with our XGboost model. Through these, we were able to conclude that the majority of feature importance is derived from fire state variables. This is a reasonable conclusion, as at a 2km hourly spatiotemporal resolution, fire spread is rare, so persistent fire pixels make up most on-fire pixels at any given time. The further channels added were each shown to have marginal gains on the model performance, and even greater gains when used by the final ConvGRU model, most likely due to the spatiotemporal capabilities.

Table 1. The 38 input channels grouped by physical category, with primary data source, spatial resolution, and temporal resolution.

Category / Channel	Source	Spatial	Temporal
<i>Fire state (3)</i>			
Previous fire footprint	GOES-16/17 ABI FDCC	2 km	Hourly
Distance to fire	Derived from footprint	2 km	Hourly
Fire-neighborhood count	Derived from footprint	2 km	Hourly
<i>Hourly weather (7)</i>			
u-wind, v-wind, gust	NOAA RTMA	2.5 km	Hourly
Temperature, dewpoint	NOAA RTMA	2.5 km	Hourly
Soil moisture	ERA5-Land	9 km	Hourly
Precipitation	GPM IMERG V07	10 km	30-min
<i>Daily fire-weather (11)</i>			
ERC, BI, FM100, FM1000	GRIDMET (Abatzoglou, 2013)	4 km	Daily
VPD, RH min/max	GRIDMET	4 km	Daily
Daily wind speed, direction	GRIDMET	4 km	Daily
LST day, LST night	MODIS MOD11A1	1 km	Daily
<i>Static terrain (9)</i>			
Slope, aspect (sin/cos), elevation, TPI	USGS 3DEP	10 m	Static
Fuel load, canopy cover	LANDFIRE	30 m	Static
Terrain ruggedness, non-burnable cover	Derived	30 m	Static
<i>Vegetation & drought (3)</i>			
NDVI, EVI	MODIS MOD13Q1	250 m	16-day
PDSI	GRIDMET Drought	4 km	Monthly
<i>Temporal encoding (4)</i>			
Hour sin/cos	Derived from timestamp	—	Hourly
Day-of-year sin/cos	Derived from timestamp	—	Daily
<i>Quality mask (1)</i>			
Validity	Derived from GOES cloud mask	2 km	Hourly

Methods

Model Architecture:

We developed FireSpreadNetV2, a custom spatiotemporal convolutional neural network (CNN) that predicts short-term wildfire spread. It combines a Convolutional Gated Recurrent Unit (ConvGRU) with a U-Net encoder–decoder and Convolutional Block Attention Module (CBAM).

We used a ConvGRU because it allows the model to build up a memory of the fire’s recent history before making its prediction. It is applied at the bottleneck of the encoder, and instead of treating each frame independently, the ConvGRU keeps a running memory across the sequence and compresses the 6 hour sliding input into one map. This ensures the final representation reflects how the fire has been evolving over the past 6 hours.

We incorporated a U-Net encoder–decoder into the CNN to ensure that the model’s predictions are informed both by local detail and broader spatial context. The encoder progressively downsamples the spatial grid while simultaneously increasing the number of feature channels, allowing the model to represent increasingly complex patterns. At higher resolutions, the model captures more local features, such as small clusters of burning pixels or local wind conditions. At lower resolutions, each feature represents a much larger spatial region, allowing the model to detect broader patterns such as advancing fire fronts or directional spread across the landscape.

The decoder then works in reverse to recover spatial resolution and produce a full-grid prediction. It progressively upsamples the compressed map back to the original grid size. At each stage, it incorporates the corresponding encoder features through skip connections. These skip

connections allow the model to combine the broader patterns (such as the overall direction of spread) with the finer details (such as the exact boundary of the fire perimeter). This structure is critical for producing sharp, pixel-level predictions rather than overly smooth outputs.

The CBAM is applied to the skip connections before they are merged with the decoder features. We included this because not all features passed through the skip connections are equally useful. CBAM applies both channel and spatial attention to reweight these features. Channel attention assigns importance scores to different feature channels based on global summary statistics, allowing the model to emphasize the most relevant learned patterns. Spatial attention assigns importance scores across locations in the grid, highlighting regions that are more informative for prediction, such as areas near active fire fronts. Together, these mechanisms help the model focus on the most relevant information when reconstructing the final prediction.

Feature Engineering and Dataset Splits:

The model accepts a multi-channel input stack of 38 variables from 11 different data sources, encoding fire state, hourly meteorology, daily fire weather indices, terrain, vegetation, and temporal features. A key feature is the inclusion of a pre-shifted fire channel, defined as $\text{prev_fire_state}[t] = \text{labels}[t-1]$. This ensures that at each timestep, the model only has access to past fire information when predicting future spread, which prevents temporal data leakage.

We constructed the feature stacks to have the shape (T, C, H, W); here, T is the number of hourly timesteps, C=38 is the number of input variables, and H x W is the spatial grid. All spatial grids were zero-padded to ensure divisibility by 16 for encoder compatibility. Training data only was

normalized per-channel by computing the mean and standard deviation, with invalid pixels excluded from these calculations using the validity mask. The validity channel itself, which is a binary flag indicating whether each pixel contains reliable data, was intentionally excluded from normalization. These training statistics were then applied directly to the test fires without recomputation, since recalculating them from test data would lead to data leakage.

We then made sure that fires with fewer than 50 total burned pixels were excluded, as events that small are insufficient for training and evaluation. The remaining fires were randomly shuffled using a fixed seed of 42 and partitioned 70/30 into training and test sets, with the Camp Fire, Creek Fire, Dolan Fire, and Glass Fire ending up in the test set. The fixed seed ensured that this same split was produced every time the code was run so that the results are reproducible.

Training Procedure:

We trained the model for 20 epochs using the AdamW optimizer with the standard learning rate of $3 * 10^{-4}$ and a weight decay of 10^{-4} . This optimizer is a step-size algorithm that adapts per-parameter and prevents the model from getting too confident in any one weight. Neural networks require massive amounts of training data, but we only had 8 training fires with around 409 hours of data for each fire. So, we trained the model on 6-hour sliding windows that advanced one hour at a time, which generated a large number of overlapping sequences from each fire. These sequences were grouped into batches of 4 before each weight update. To stabilize training even more, we clipped the gradient norms to a maximum of 1.0 at each step, preventing any single bad batch from causing an erroneously large weight update.

We also applied data augmentation during training through the function `augment_sequence_v2`. This function flipped our fire maps horizontally, vertically, or both, and fed the flipped map into the model as “new” training data; this worked because the modified copies are still physically valid fires, but the model had never seen that exact version before, so it counted as new training data. We had to be sure to flip the input wind variable while doing this, since the wind is a directional input; so, when we flipped the map horizontally, for example, we flipped the east-west wind component as well so that the samples remained physically consistent.

Loss Function:

Our final total training loss function combines a supervised data term with a physics-informed regularization term:

$$L = L_{data} + \lambda * L_{physics}$$

The data loss is an equally weighted combination at 0.5 each of Focal Loss and Focal Tversky Loss. We used Focal Loss because wildfire training data has a severe class imbalance - burned pixels are vastly outnumbered by unburned pixels. Focal Loss accounts for this by downweighting easy negatives, causing the model to give less attention to the pixels it is very confident are unburned, and more attention to the rarer fire pixels. We used Focal Tversky Loss because it penalizes false negatives more heavily than false positives, which is desirable because it is more dangerous to not predict a fire will spread when it does, than to predict it will spread when it does not. This asymmetry is controlled by setting the false-positive weight α to 0.3 and the false-negative weight β to 0.7. An additional focal exponent of $\gamma = 0.75$ is applied on top of the Tversky index to further focus on hard examples. This causes pixels where the model is

already doing reasonably well to get their loss contribution shrunk down, while pixels where the model is doing very badly have a higher loss contribution.

The physics loss encodes a wind advection constraint derived from the following PDE:

$$\frac{\delta F}{\delta t} = -u \frac{\delta F}{\delta x} - v \frac{\delta F}{\delta y}$$

where F is the fire-state and (u, v) are the horizontal and vertical wind components, respectively. At each training step, spatial gradients of the previous fire state are computed using Sobel filters, and the expected wind-driven fire displacement is calculated. The Sobel filters discretize the spatial derivatives onto our pixel grid using a weighted center-difference scheme across neighboring rows and columns, which is normalized by a factor of 8 in order to scale the gradient estimates appropriately. The expected wind-driven fire displacement is then calculated from these gradients and the wind components. The physics penalty is the mean squared residual between the model's predicted change in fire state, and the change predicted by pure wind advection alone, which discourages the model from predicting spread that is inconsistent with observed wind direction and magnitude. The weighting term λ controls the contribution of the physics penalty to the total loss signal. It was set to 0.2 after experimental tuning, which found that lower values barely impacted F2 scores and higher values actually decreased the F2 score due to overfitting to the PDE.

Previous Models

We arrived at our final architecture after a sequence of experiments spanning three model classes: per-pixel logistic regression, XGBoost, and finally, the ConvGRU U-Net.

Our first two experiments, the per-pixel logistic regression and XGBoost models, relied on the surrounding 3x3 neighborhood of pixels to predict the center at the next timestep, differing from our ConvGRU U-Net’s image prediction output. Initially, these models had F1 scores of 0.72-0.75, with our ConvGRU U-Net being the strongest.

A further analysis of the data found that clouds, as well as smoke from the fires, was obscuring pixels, and preventing the satellite data from correctly identifying them as on fire. To fix this, we applied a five-frame majority-vote filter, effectively smoothing the pixels that were on fire and suddenly “blinked out” for one frame due to smoke or clouds. This, as well as further general cleaning of our dataset, resulted in significant improvements in model performance, and inspired us to explore further data-based improvements, such as the physics informed constraints.

Results

How Results are Measured

We wanted to measure both false negative and false positive rates which lead us to using f scores. We first used F1 score which equally weights both recall¹ and precision². While low precision is not ideal and can lead to misdirected resources that could have been vital in other areas, low recall is worse. Low recall would mean fires being unpredicted leading to large scale damage and potential for deaths. For this reason we switched using F2 score³ which weighs recall twice as much as precision.

Aggregate and Per-Fire Performance

We trained our final model on eight California wildfires (3,560 hours, 2.9 million acres) and tested on four held out fires (1,809 hours, 726,000 acres): the Camp Fire (2018), and the Creek, Dolan, and Glass fires (all 2020). Across the held-out set, the ConvGRU U-Net model achieved an aggregate F2 score over the test fires of 0.928, with precision 0.922, and recall of 0.930. Per-fire performance was stable across the four test events, with F2 ranging from 0.918 to 0.933.

In aggregate, recall (0.930) exceeded precision (0.922), which was consistent with the F2 objective, influenced by the false-negative-heavy Focal-Tversky loss used during training. This follows our intended priority of identifying fire spread over the accuracy of our predictions.

Table 2. Per-fire confusion-matrix counts and F2 score on the held-out test set.
 TP = true positives, FN = false negatives, FP = false positives, TN = true negatives.

Fire	Year	TP	FN	FP	TN	F2
Camp	2018	8,067	635	615	53,616	0.933
Creek	2020	42,669	3,123	3,413	632,355	0.927
Dolan	2020	10,100	779	923	237,838	0.918
Glass	2020	5,462	458	631	87,600	0.926
Aggregate		66,298	4,995	5,582	1,011,409	0.928

Architectural and Data Pipeline Progression

As we developed our model architecture and data pipeline, we trained the three models: per-pixel logistic regression, XGBoost, and the ConvGRU U-Net, on both the pre-corrected pipeline and after. Table 3 shows the progression of F1, as well as final F2 score after the pipeline fix.

Table 3. Performance of three architectures across two pipeline generations.
 The two F1 columns differ only in the labels and feature-construction pipeline;
 no architectural or hyperparameter changes were made between them.

Model	F1, original	F1, corrected	Δ F1	F2, corrected
Logistic regression	0.716	0.832	+0.116	0.900
XGBoost	0.744	0.884	+0.140	0.926
ConvGRU-U-Net	0.749	0.929	+0.180	0.928

Comparison with Published Models

Table 4 shows our model in comparison to four references for wildfire spread prediction. No published benchmark reports F2 as its primary evaluation metric, and no ML-based prediction benchmark operates at hourly resolution. Therefore, a direct F1/performance comparison is not well-defined.

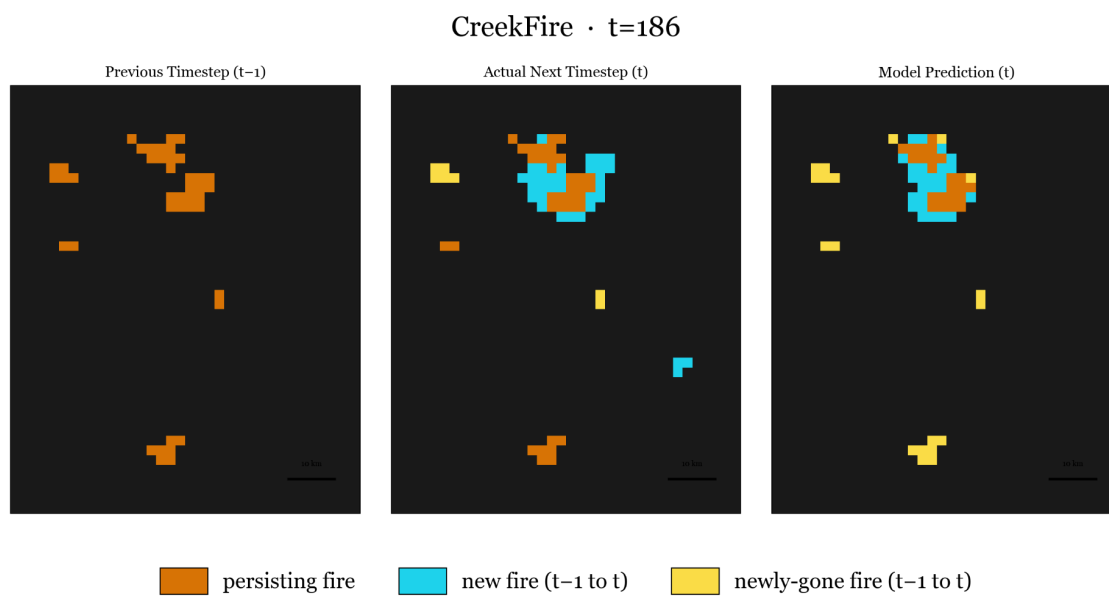
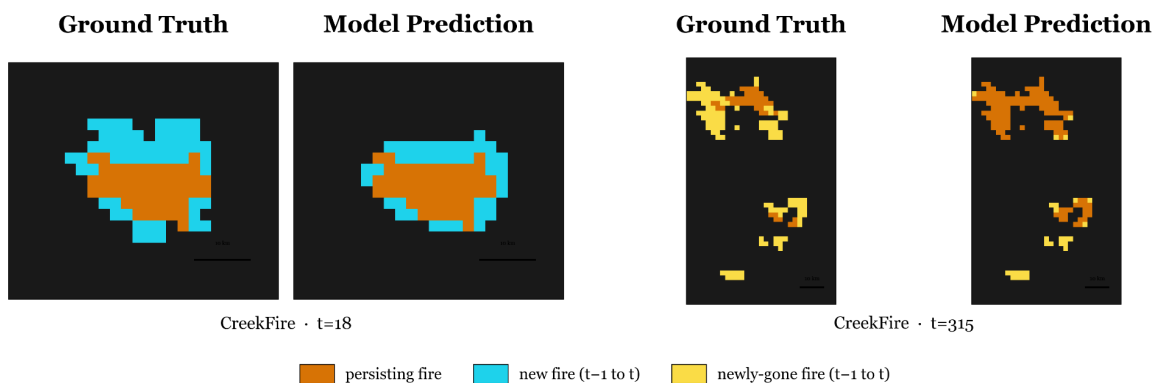
Table 4. Position relative to published references for wildfire spread prediction and fire-state observation.

Reference	Spatial	Temporal	Source	Type
NDWS (Huot et al., 2022)	1 km	Daily	MODIS/VIIRS	ML benchmark
WildfireSpreadTS (Gerard et al., 2023)	375 m	Daily	VIIRS	ML benchmark
WSTS+ (2025)	375 m	Daily	VIIRS	ML benchmark
GOFER (Artés et al., 2024)	~ 2 km	Hourly	GOES	Perimeter retrieval
This work	~ 2 km	Hourly	GOES	ML spread predictor

GOFER is similarly a GOES-based hourly fire-perimeter prediction model, but relies on physics-based methods, and uses vector based borders.

Qualitative Results

The figures below show ground-truth and predicted fire-state matrices for the Creek Fire at three timesteps: an early-phase spreading ($t = 18$ hours after ignition) a later-phase retreat ($t = 315$ hours) and finally a more complex scenario with both spread and retreat of fire-front ($t = 186$). Boundary errors are largely localized to within one pixel of truth for all predictions.



Conclusion

This work presents FireSpreadNetV2, a spatiotemporal convolutional neural network that predicts hourly wildfire spread at 2km resolution. By operating on hourly scales, the model addresses a limitation of prior machine learning approaches that were limited to daily resolutions. Trained on eight California wildfires and evaluated on four held-out fires, the model achieves an aggregate F2 score of 0.928, with recall exceeding precision in a manner consistent with our training objective. Qualitative results further indicate that the model captures fire perimeter dynamics with meaningful spatial precision.

There are several limitations within the model that provide opportunities for future work to address. The resolution of 2km is sufficient for regional spread prediction, but coarse for fine-scale operational decisions; more spatially detailed data sources could be of benefit, as the model could then provide predictions at those finer scales. Also, the model was trained and evaluated exclusively on California wildfires, which limits its generalizability to other fuel types and climates. Training in other areas outside of California would increase the global generalizability of the model.

Despite these limitations, the results demonstrate that hourly ML-based fire spread prediction can be used to improve the accuracy of fire prediction in the future. This is significant because local governments can implement it to understand where fires will spread and for how long they will burn. This will allow them to issue appropriate evacuation orders to regions under threat; to have efficient resource allocation to the most heavily affected areas; and to be able to proactively combat wildfire spread by allowing them to understand the movement of the fire front.

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Appendix

Equations:

$$1. \text{ Recall} = \frac{\text{True Positives}}{\text{Real Positives}}$$

$$2. \text{ Precision} = \frac{\text{True Positives}}{\text{Predicted Positives}}$$

$$3. \text{ F2 score is } F_{\beta} = \frac{(1+\beta^2)}{\beta^2} * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}} \text{ where } \beta = 2$$

Github Repository:

<https://github.com/seanpatrickmay/wildfire-modeling.git>